

LecW04A Monday: Asynchronous. Recording's link will be in Brightspace. I'll be traveling for a conference.

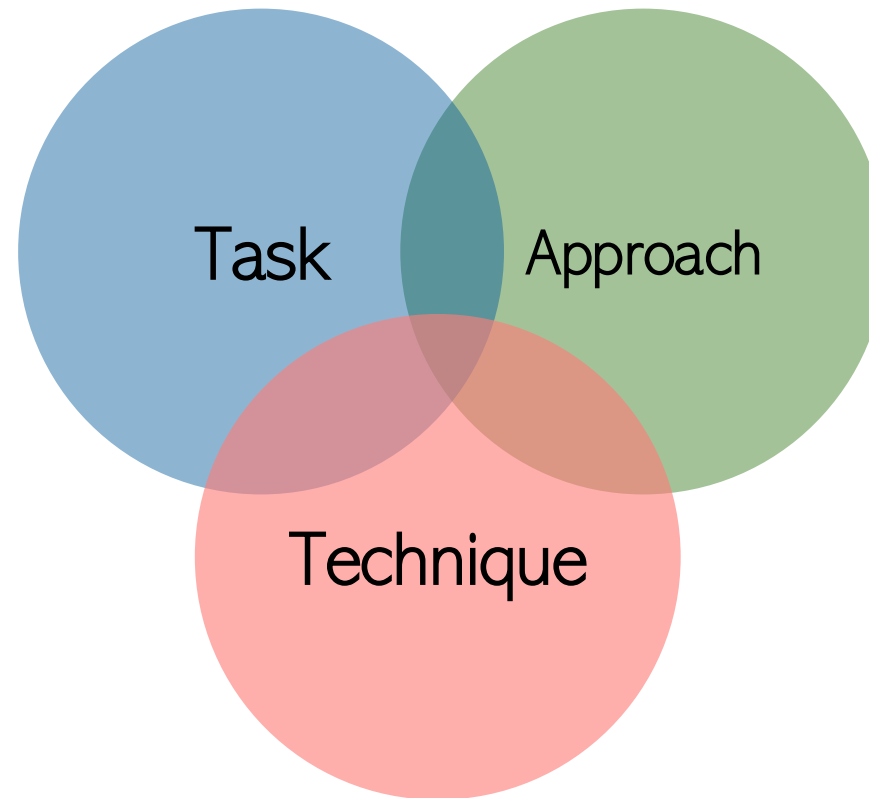


An image of the core region of a supermassive black hole, processed from a sparse array of radio telescopes
https://en.wikipedia.org/wiki/Black_hole

Outline: W02B: Topics Start

Classification

Regression
Clustering
Ranking (Recommendation)
Vector Representation and
Dimensionality Reduction
Language Modeling



Full/Self/Semi-Supervised Learning

Unsupervised Learning
Reinforcement Learning

k-Nearest Neighbor, Naive Bayes, Logistic Regression
Artificial Neural Networks, Graph Neural Network
Support Vector Machines
Principal Component Analysis
K-Means

4 Outline: W03A: Topics Start

- I) Task: Classification
- II) Supervision: Fully Supervised
- III) Technique: k -Nearest Neighbor
- III) Evaluation

5

k -Nearest Neighbor: Natural Language

"We become who we hang out with" - Napoleon Hill

"If you hang around 5 confident/intelligent/millionaires/idiots, you will be the 6th – Denzel Washington



Oliver Napoleon Hill (1883 –1970)

American self-help author and **conman**.

Known for his book *Think and Grow Rich* (1937), the best-selling self-help books of all time.

Most of his books are promoted as expounding principles to achieve "success".

A controversial figure, **history of criminal charges and fraud accusations**, historians also doubt many of his claims.

Two Academy Awards, an Actor Award, two Golden Globes, two Silver Bears and a Tony Award as well as nominations for a Grammy Award and two Primetime Emmy Awards.



Denzel Hayes Washington Jr. (1954-Present)

American actor and producer.

Known for his dramatic roles on stage and screen.

6 Outline: W03A: Topics Start

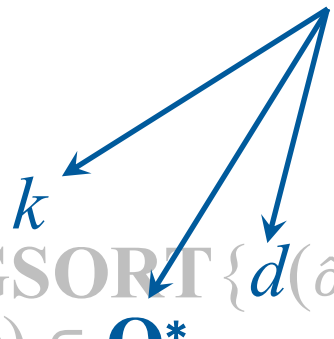
- I) Task: Classification
- II) Supervision: Fully Supervised
- III) Technique: k -Nearest Neighbor
Let's zoom in ... Hyperparameters
- III) Evaluation



Tug of war (also known as tug o' war, tug-a-war, tug war, rope war, rope pulling, or tugging war)

8

k -Nearest Neighbor: Let's zoom in ...

$$f(\hat{o} : \mathbf{O}^*, k, d) = \underset{c_i \in \mathbf{C}}{\text{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \text{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}} \mathbf{1} [c_i = c] \right\}$$


No internal memory (parameters) $\rightarrow$ *non*-Parametric (no summarization)

But it has Hyperparameters

$\rightarrow$ Compile-time parameters

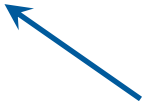
Form the learning algorithm architecture (structure)

Are “*tuned*” during design to find the best architecture

Remain constant after deploy for production

9

k -Nearest Neighbor: Efficacy vs. Efficiency Trade-Off

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$


Scaling labeled dataset:

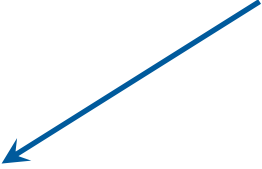
→ Small → Fast but **Poor**

→ Large → **Slow** but Accurate

Labeled set of observations (training set) is commonly not considered a hyperparameter. Why?

10

k -Nearest Neighbor: Variance vs. Bias Trade-Off

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$


k , number of nearest neighbor

→ Small like $k = 1$

Liberal: decides based on whatever sits next to $\hat{o}$

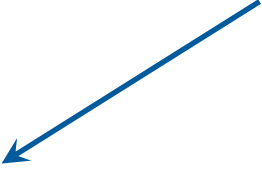
Unbiased (Low Bias): not impose a strong global preference

Locally Sensitive

High Variance

11

k -Nearest Neighbor: Variance vs. Bias Trade-Off

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$


k , number of nearest neighbor

→ Large like $k = |\mathbf{O}^*|$

Conservative: considers a broader neighborhood

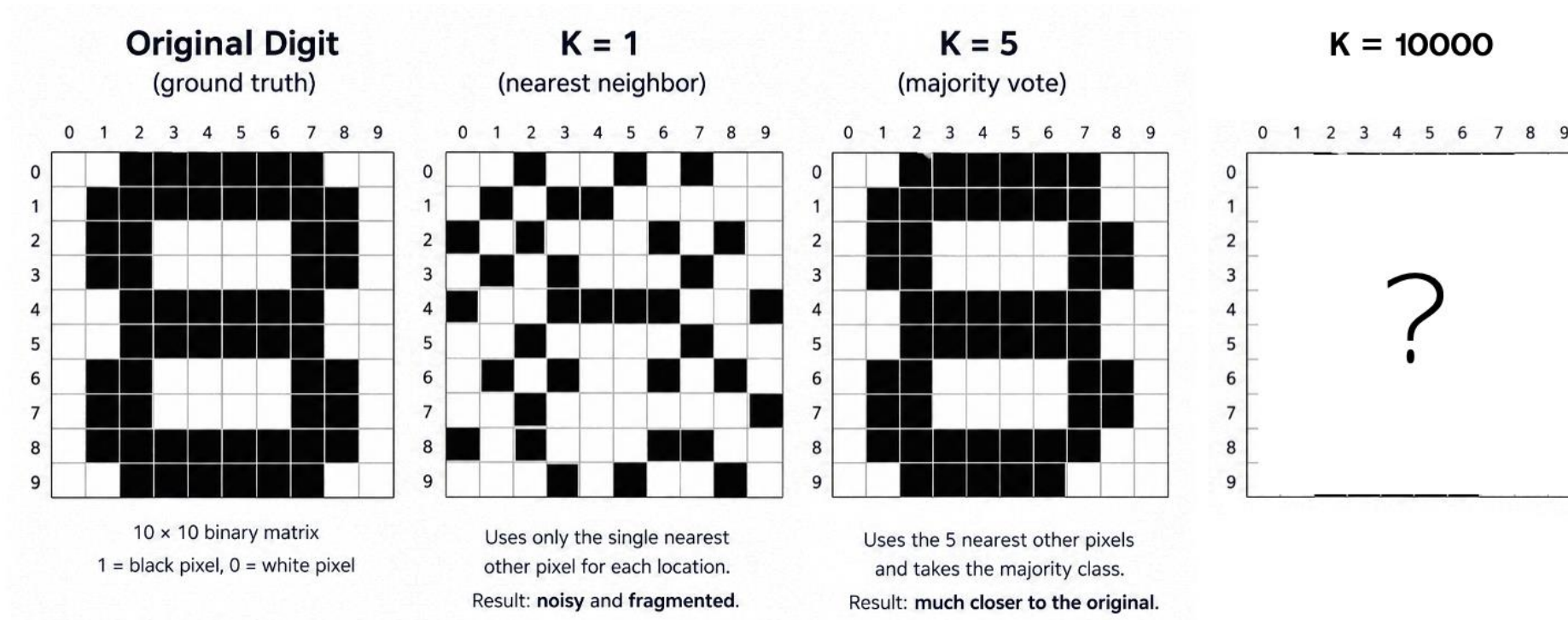
Low Variance

Biased (High Bias): Blind, favors the global over local pattern

Globally Conforming, Another Brick in the Wall :|

k -Nearest Neighbor: Inference on Training Set

A good learner (technique) can easily learn the training set.
A student whose exam contains only the lecture contents.



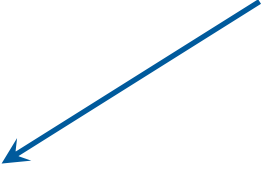
We make a copy of an image, then restore every pixel of it by its surrounding ...



Daniel Edmond Sullivan Jr
Ocean Photographer of the Year
<https://www.nationalgeographic.pt/fotografia/imagens-que-mostram-beleza-oculta-mares-e-oceanos-concurso-ocean-photographer-of-the-year> 7414

14

 r -Nearest Neighbor: Neighbors vs. Neighborhood

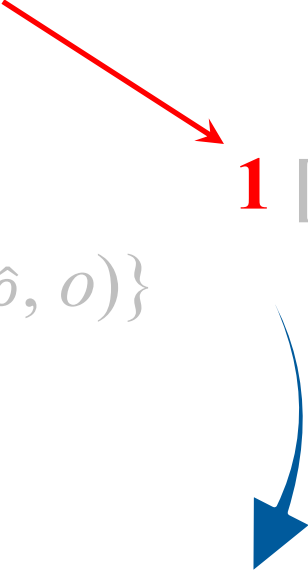
$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}} \underset{d \leq r}{1} [c_i = c] \right\}$$


$k \in \mathbb{N}^+$ number of nearest neighbors $\rightarrow r \in \mathbb{R}^{\geq 0}$ radius of nearest neighborhood

- $\rightarrow$ Small, like r close to 0 $\rightarrow$ Empty neighborhood, cannot decide.
- $\rightarrow$ Large, like $r = +\infty \rightarrow$ Very crowded neighborhood, conformity.

15

 k -Nearest Neighbor: Weighted (“Attentive”)

$$f(\hat{o} : \mathbf{O}^*) = \operatorname{ARGMAX} \left\{ \sum_{c_i \in \mathbf{C}} \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k \mathbf{1} [c_i = c] \right\}$$


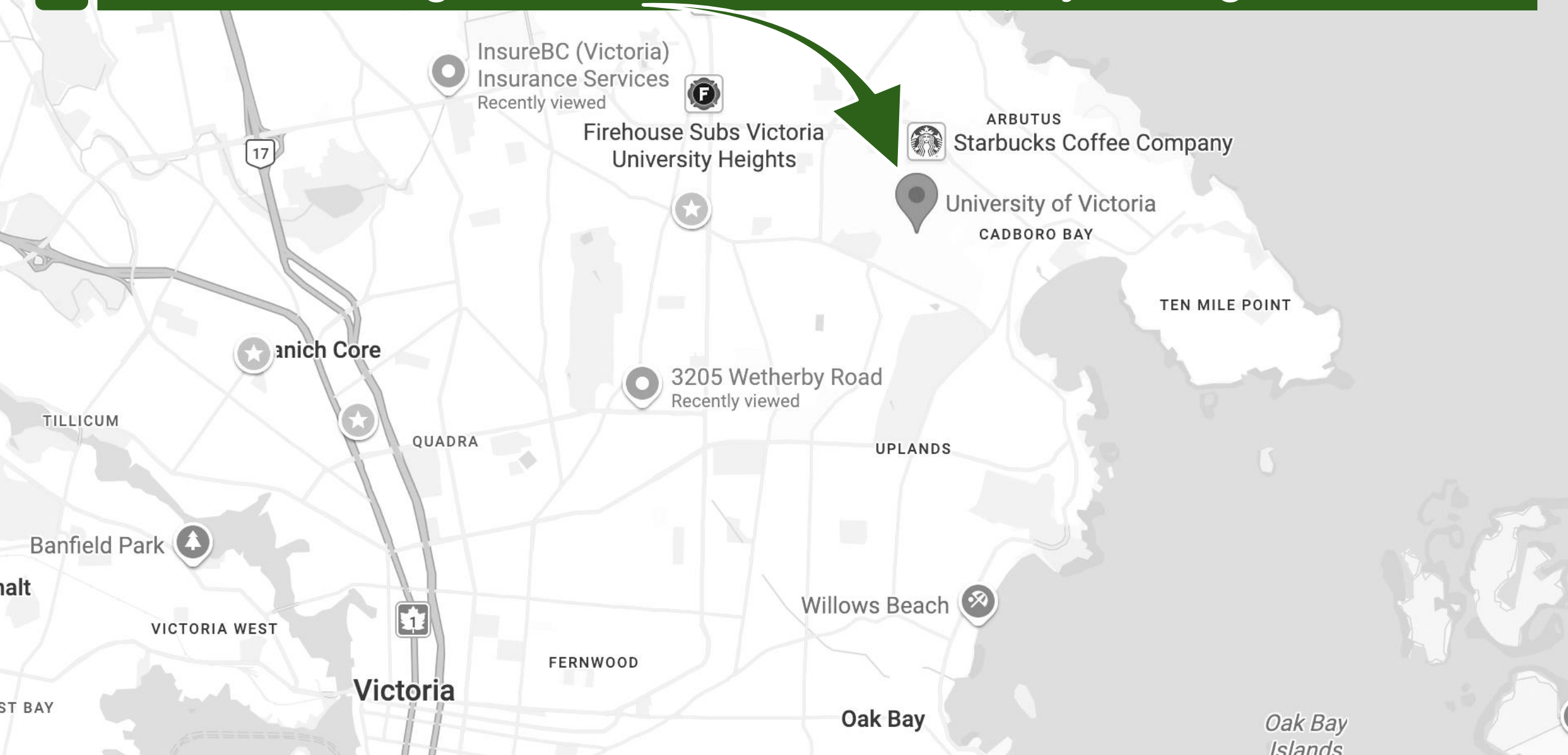
$$f(\hat{o} : \mathbf{O}^*) = \operatorname{ARGMAX} \left\{ \sum_{c_i \in \mathbf{C}} \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k \frac{1}{d(\hat{o}, o)} \times [c_i = c] \right\}$$

16

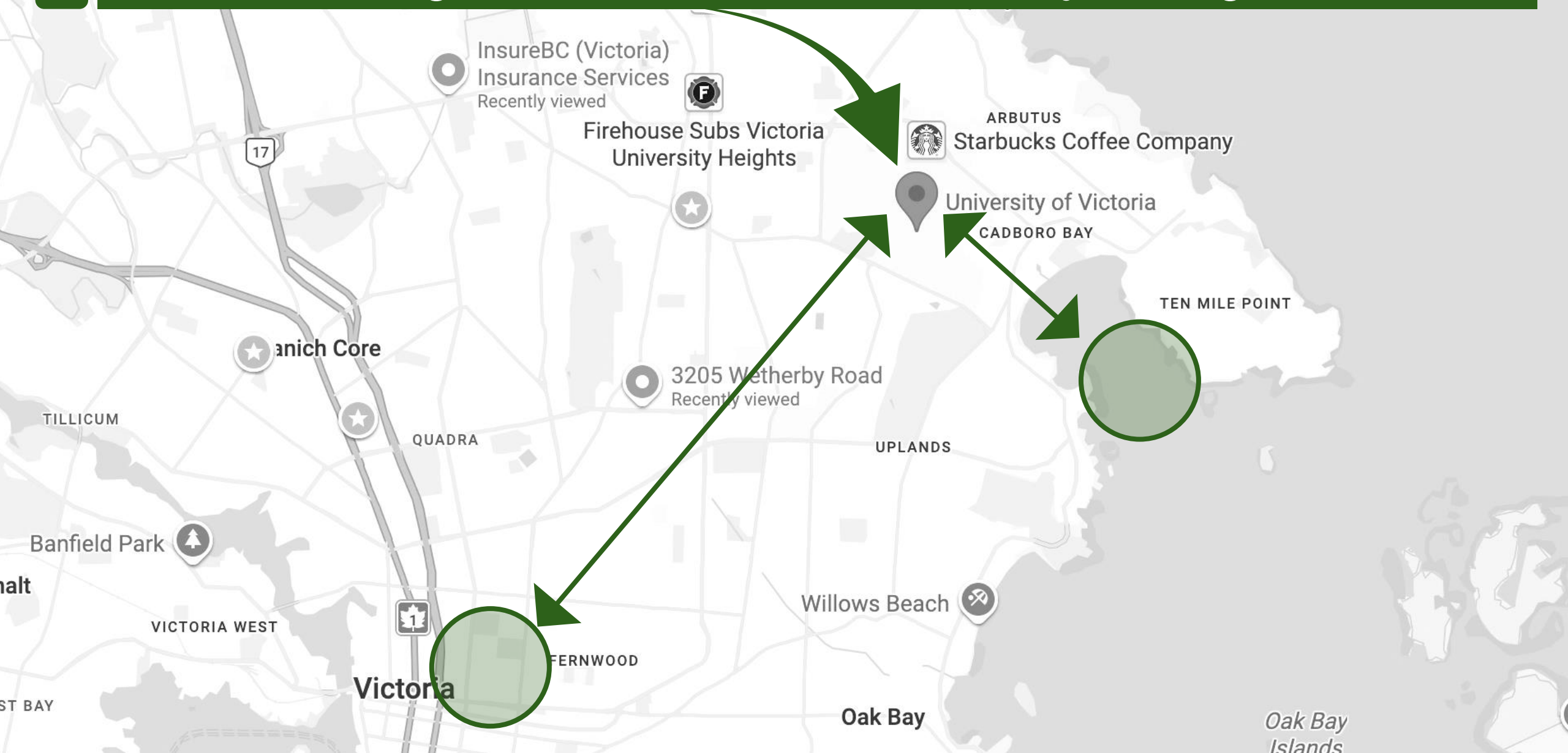
 k -Nearest Neighbor: Distance Function

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$

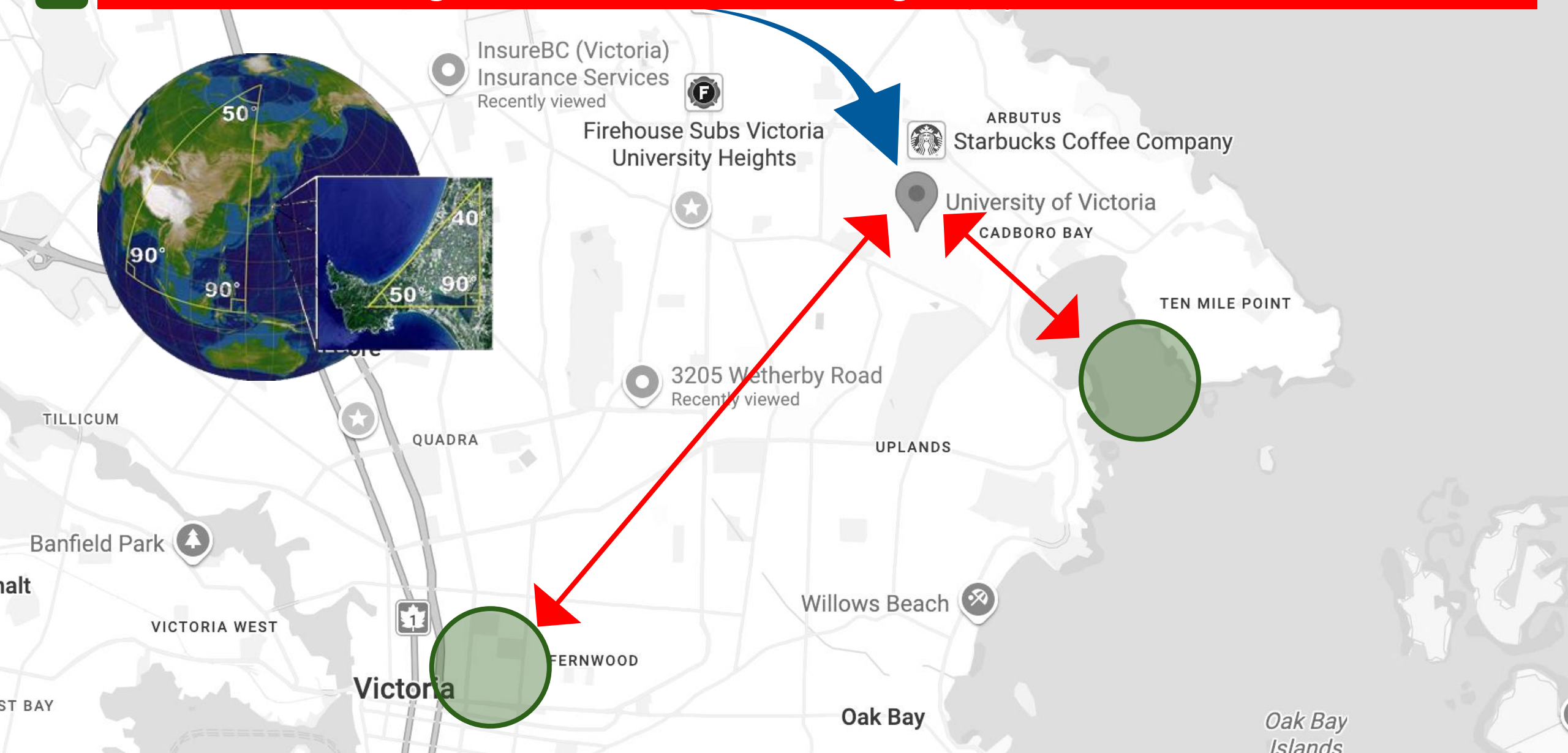

k -Nearest Neighbor: Traffic at UVic? Heavy vs. Light



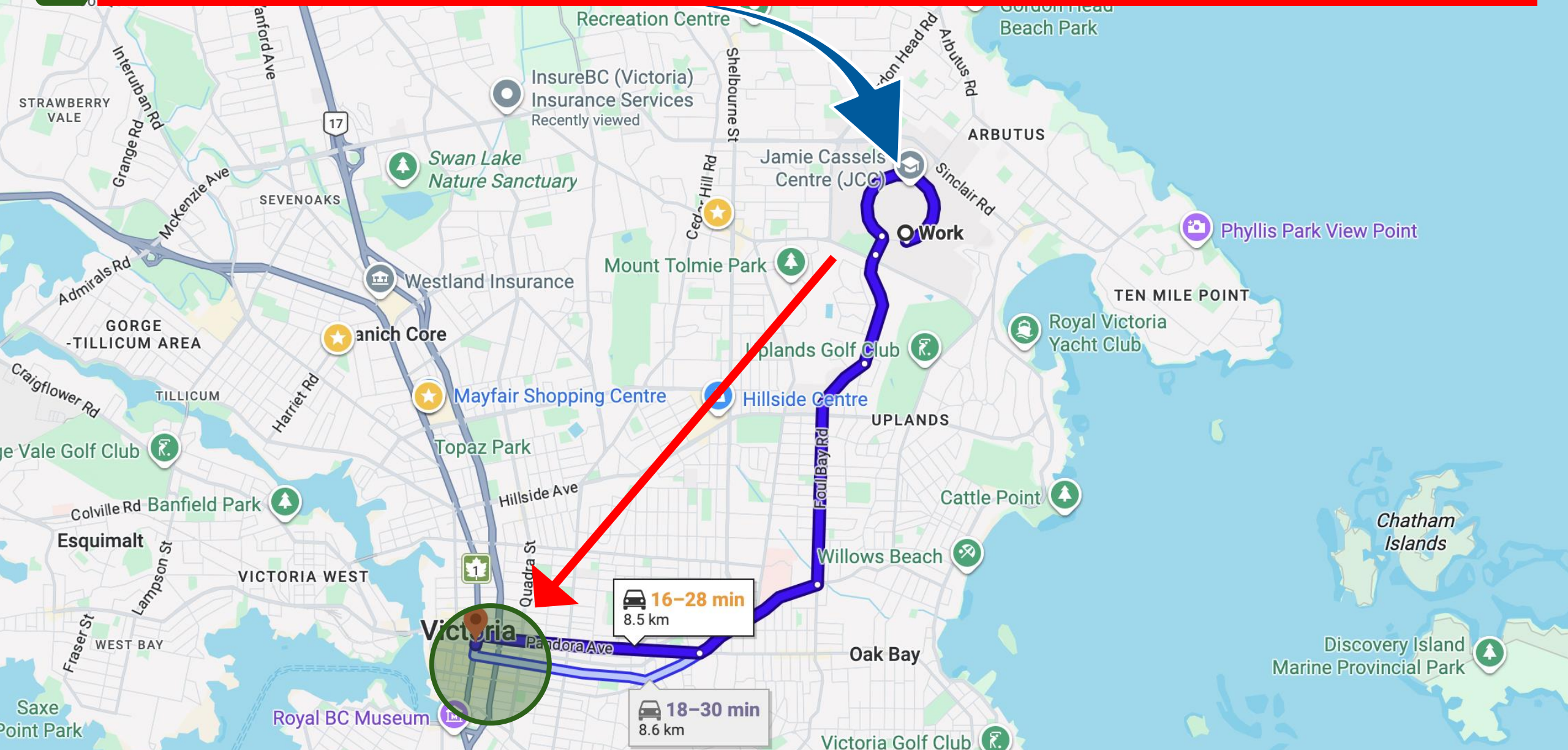
k -Nearest Neighbor: Traffic at UVic? Heavy vs. Light



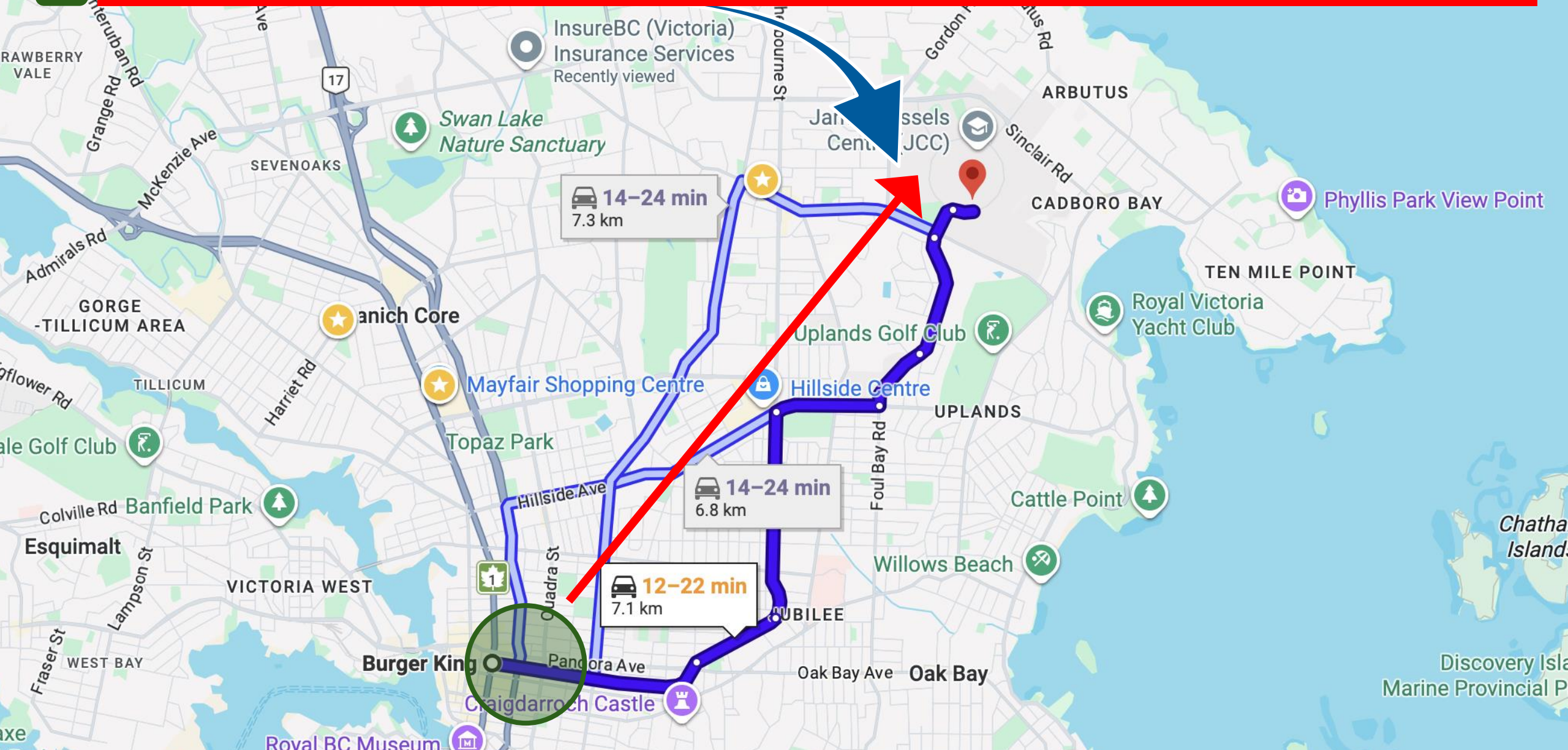
k -Nearest Neighbor: Math Challenges



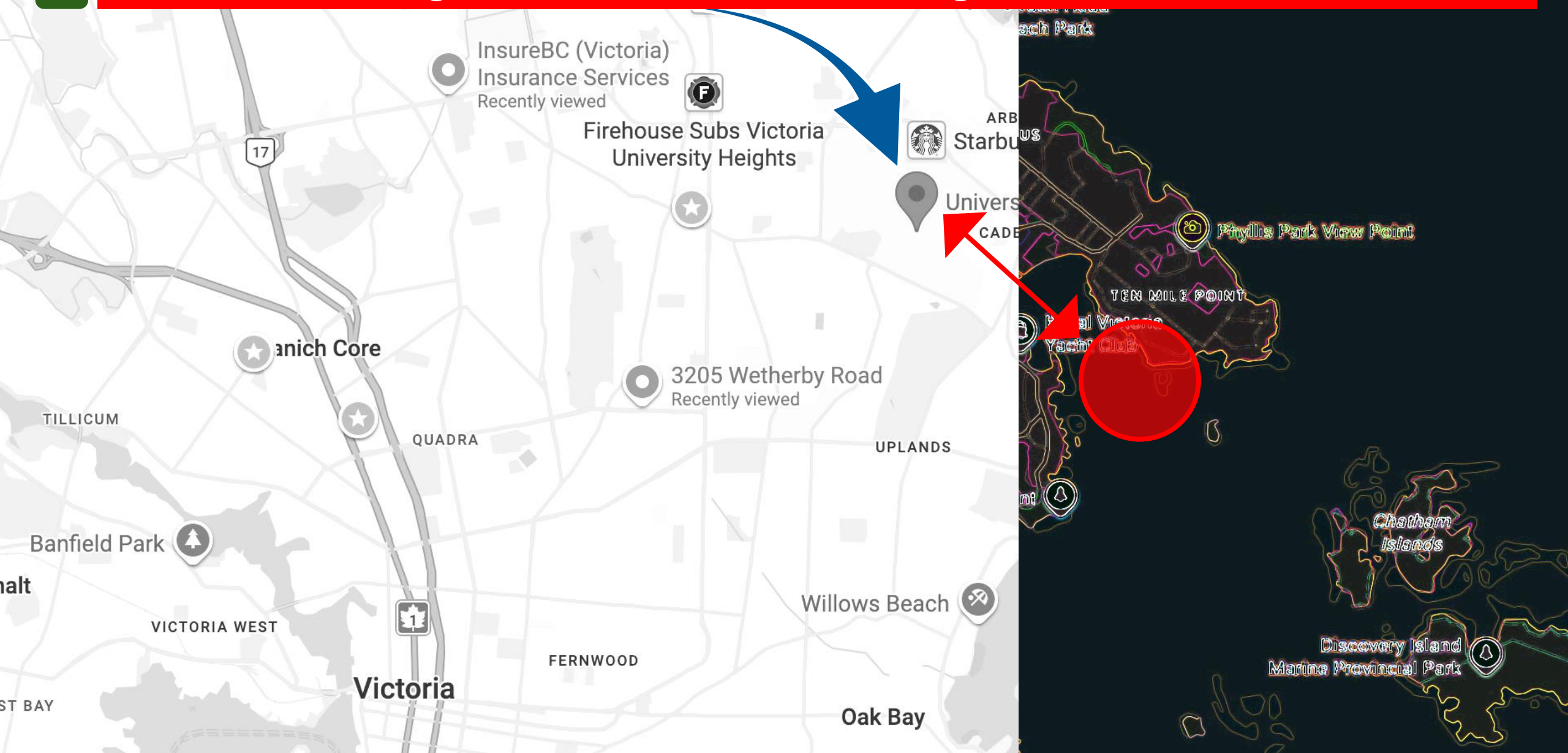
k -Nearest Neighbor: Domain Challenges



k -Nearest Neighbor: Domain Challenges



k -Nearest Neighbor: Domain Challenges



23

k -Nearest Neighbor: Space Hierarchy

We need a space that covers observations for defining distance function.

More restrictions

More assumptions

More structure

More idealized

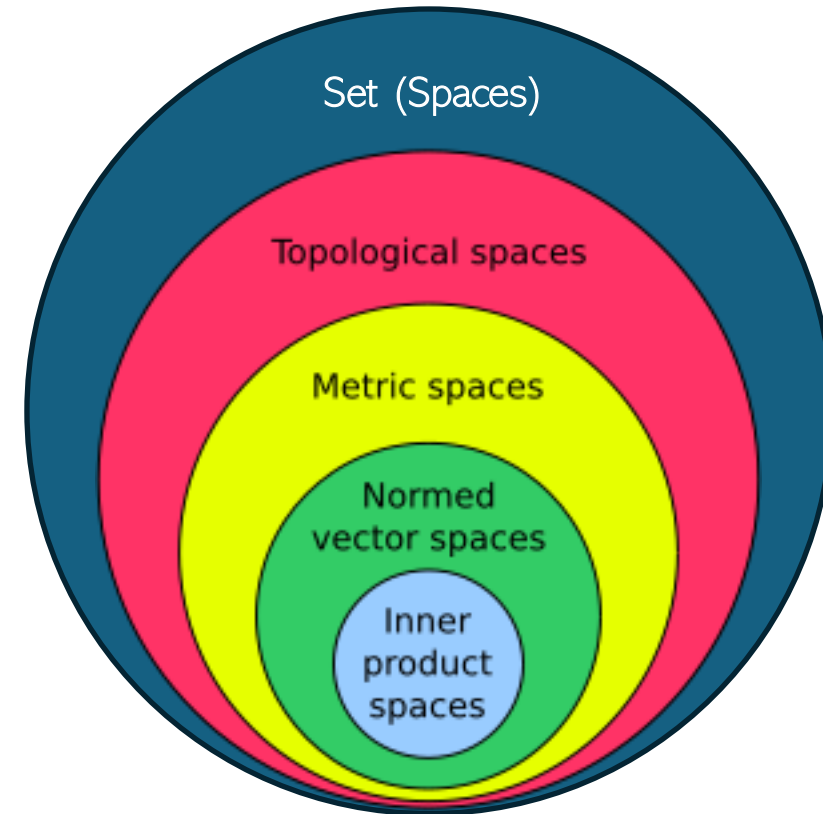
Less realized

Less general

Easier mathematics

Easier to work with

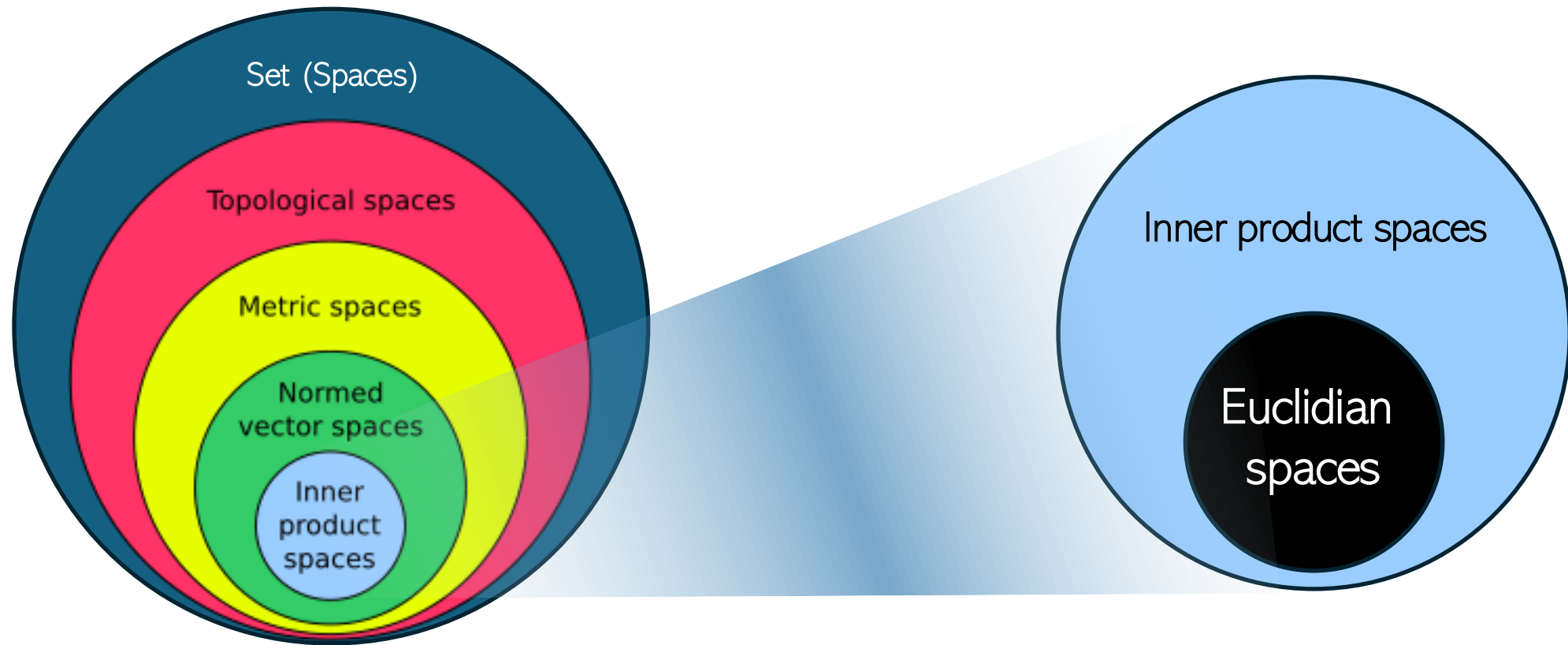
“give up generality to gain tractability”



24

k -Nearest Neighbor: Space Hierarchy

We need a space that covers observations for defining distance function.



“nice geometric space”

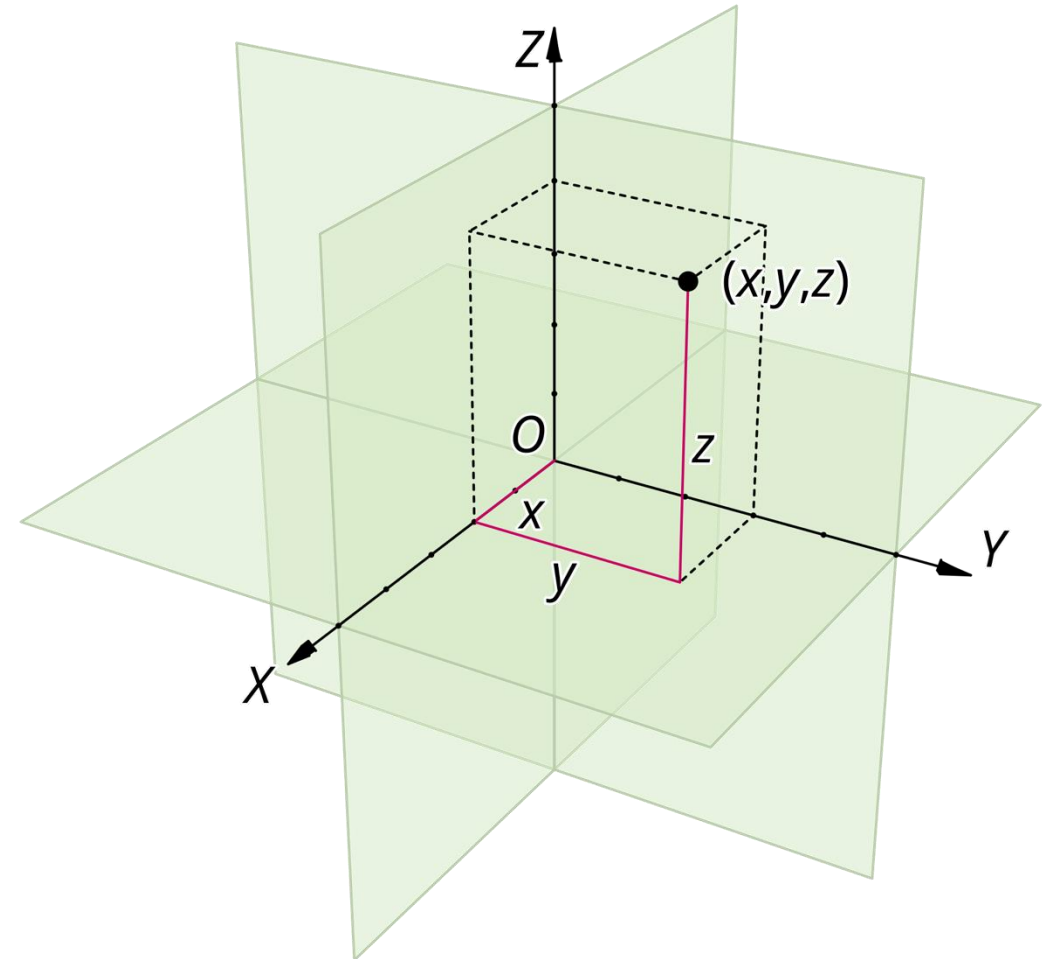
yoo-KLID-ee-uhn

k -Nearest Neighbor: Euclidian Space



Detail of Raphael's impression of Euclid, teaching students in *The School of Athens* (1509–1511)

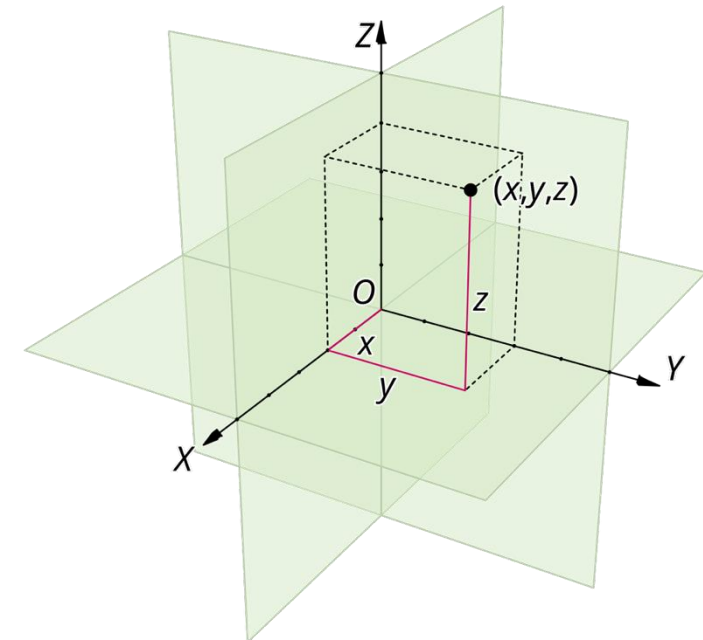
Euclid (/ˈjuːklɪd/; Ancient Greek: Εὐκλείδης; fl. 300 BC)



k -Nearest Neighbor: Euclidian Space

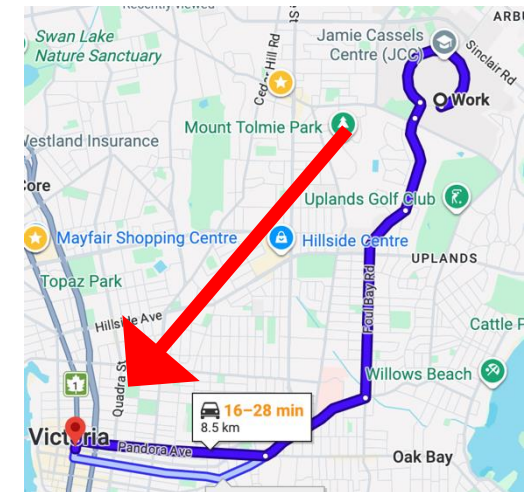
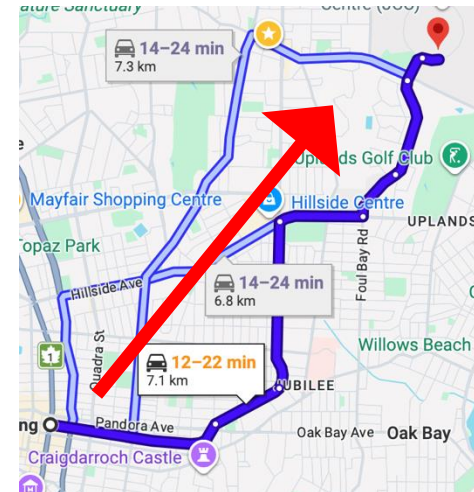
If $\mathbf{p}$ and $\mathbf{q}$ are two “points” (pls don't use the term “vector” for now):

1. $d(\mathbf{p}, \mathbf{q}) \geq 0.0$ (no negative distances).
2. $d(\mathbf{p}, \mathbf{q}) = 0.0 \Leftrightarrow_{\text{iff}} \mathbf{p} = \mathbf{q}$.
3. $d(\mathbf{p}, \mathbf{q}) = d(\mathbf{q}, \mathbf{p})$ (symmetric).
4. $d(\mathbf{p}, \mathbf{q}) \leq d(\mathbf{p}, \mathbf{x}) + d(\mathbf{x}, \mathbf{q})$ (triangle inequality).



k -Nearest Neighbor: *non*-Euclidian Space

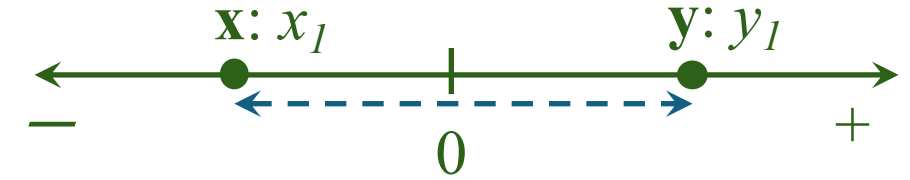
1. $d(\mathbf{p}, \mathbf{q}) \geq 0$ (no negative distances).
2. $d(\mathbf{p}, \mathbf{q}) = 0 \iff_{\text{iff}} \mathbf{p} = \mathbf{q}$.
3. $d(\mathbf{p}, \mathbf{q}) \neq d(\mathbf{q}, \mathbf{p})$ (symmetric).
4. $d(\mathbf{p}, \mathbf{q}) \not\leq d(\mathbf{p}, \mathbf{x}) + d(\mathbf{x}, \mathbf{q})$ (triangle inequality).
 $\geq$



k -Nearest Neighbor: Euclidian Space

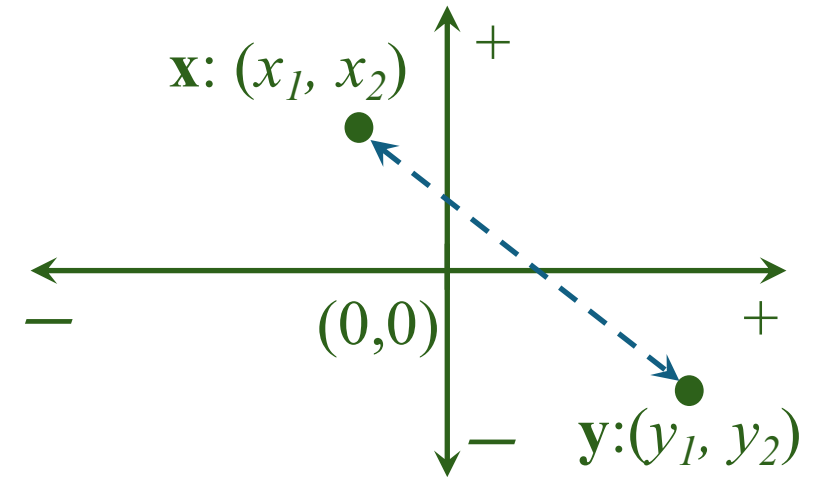
1 D Euclidian Distance $\mathbb{R}^1 \rightarrow \mathbb{R}$

$$\begin{aligned}d(x_1, y_1) &= [(x_1 - y_1)^2]^{1/2} \\&= [\sum_{i \in \{1\}} (x_i - y_i)^2]^{1/2} \\&= |x_1 - y_1|\end{aligned}$$



2D Euclidian Distance $\mathbb{R}^2 \rightarrow \mathbb{R}$

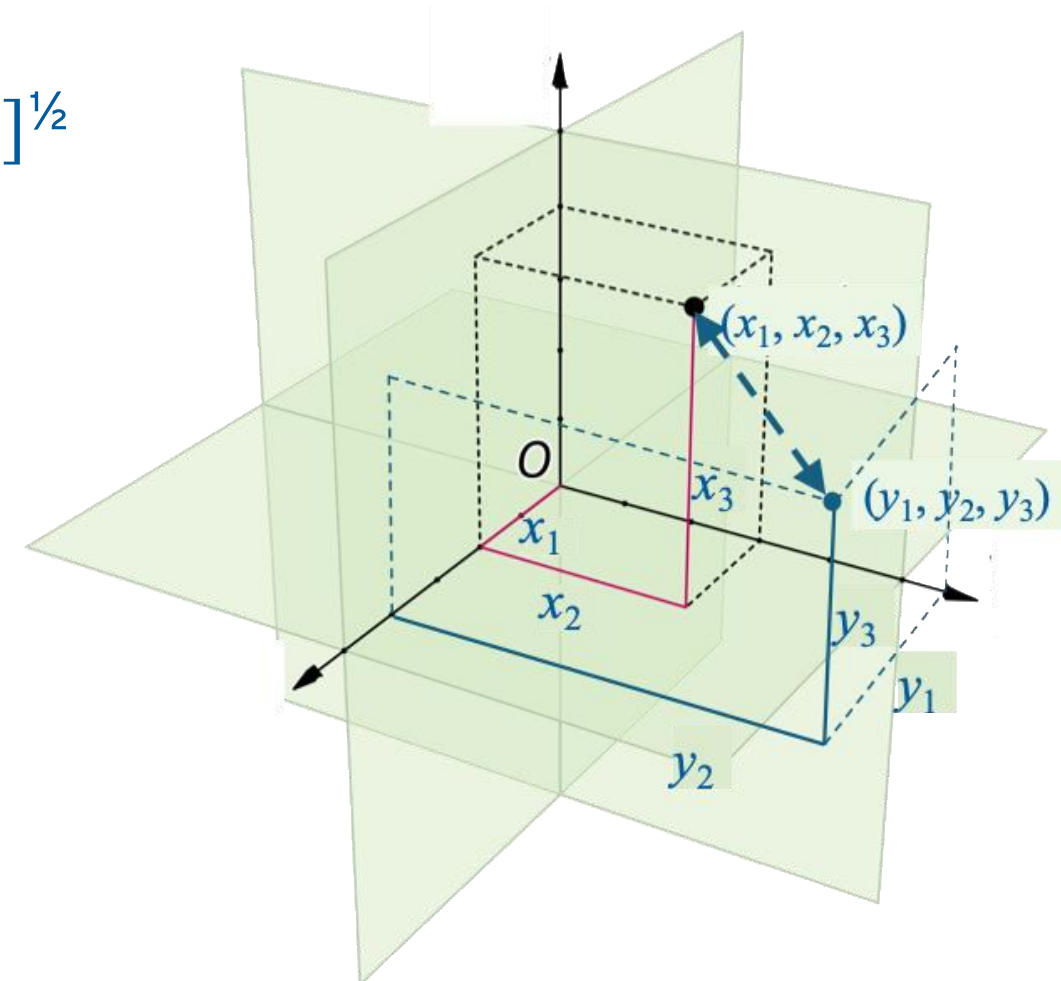
$$\begin{aligned}d(\mathbf{x}, \mathbf{y}) &= d((x_1, x_2), (y_1, y_2)) \\&= [(x_1 - y_1)^2 + (x_2 - y_2)^2]^{1/2} \\&= [\sum_{i \in \{1,2\}} (x_i - y_i)^2]^{1/2}\end{aligned}$$



k -Nearest Neighbor: Euclidian Space

3D Euclidian Distance $\mathbb{R}^3 \rightarrow \mathbb{R}$

$$\begin{aligned} d(\mathbf{x}, \mathbf{y}) &= d((x_1, x_2, x_3), (y_1, y_2, y_3)) \\ &= [(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2]^{\frac{1}{2}} \\ &= [\sum_{i \in \{1,2,3\}} (x_i - y_i)^2]^{\frac{1}{2}} \end{aligned}$$



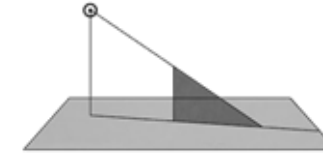
30

k-Nearest Neighbor: Euclidian Space

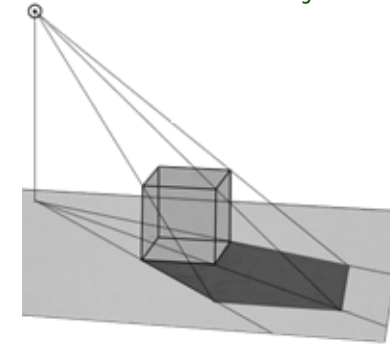
4D Euclidian Distance $\mathbb{R}^4 \rightarrow \mathbb{R}$

$$\begin{aligned} d(\mathbf{x}, \mathbf{y}) &= d((x_1, x_2, x_3, x_4), (y_1, y_2, y_3, y_4)) \\ &= [\sum_{i \in \{1,2,3,4\}} (x_i - y_i)^2]^{\frac{1}{2}} \end{aligned}$$

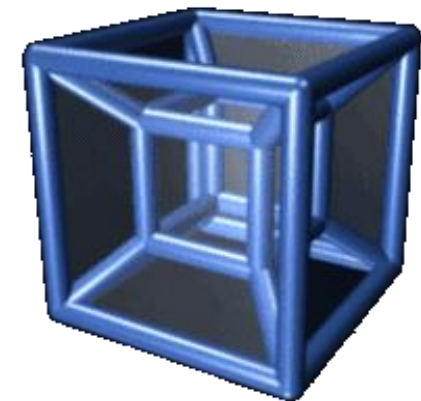
Shadow of a 2D object becomes 1D



Shadow of a 3D object becomes 2D



Shadow of a 4D object becomes 3D



A tesseract or 4-cube

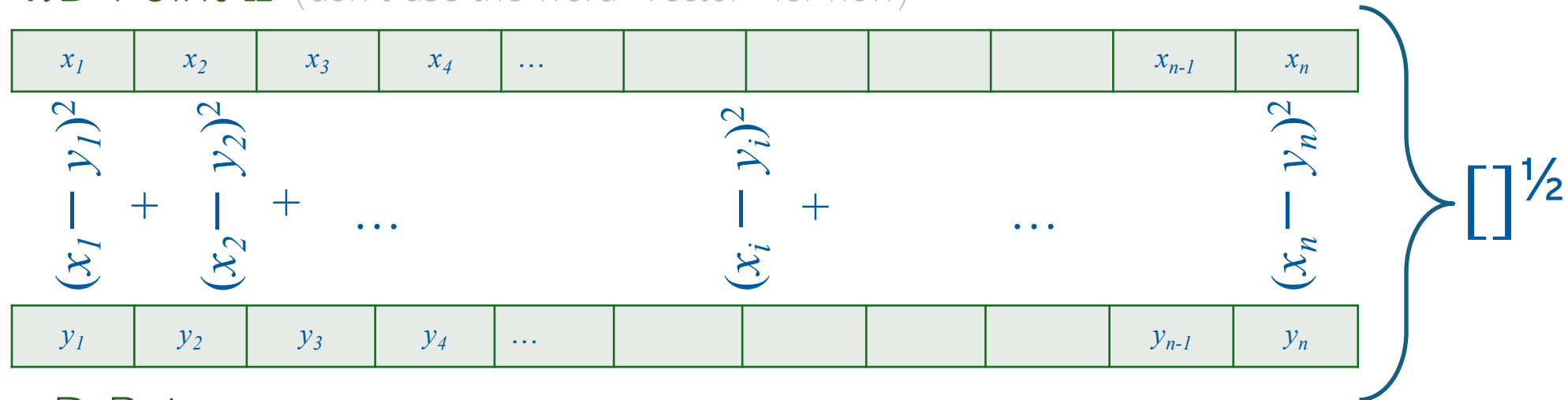
31

k -Nearest Neighbor: Euclidian Space

n D Euclidian Distance $\mathbb{R}^n \rightarrow \mathbb{R}$

$$d(\mathbf{x}, \mathbf{y}) = d((x_1, \dots, x_n), (y_1, \dots, y_n)) = [\sum_{i \in \{1, \dots, n\}} (x_i - y_i)^2]^{1/2}$$

n D Point $\mathbf{x}$ (don't use the word "vector" for now)



n D Point $\mathbf{y}$ (don't use the word "vector" for now)

k -Nearest Neighbor: Minkowski Distance



Minkowski in 1883, at the time of being awarded the Mathematics Prize of the French Academy of Sciences

$$\left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}$$

$p = 1$; Manhattan Distance
 $p = 2$; Euclidean Distance
 $p = +\infty$; Chebyshev Distance

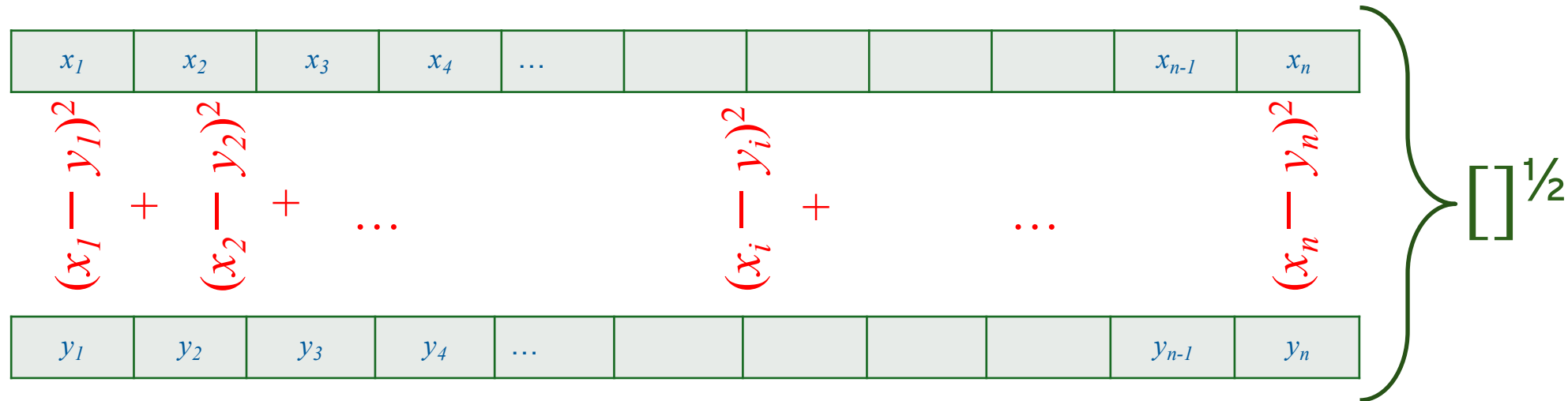
Hermann Minkowski (1864 – 1909)

Mathematician & professor at University of Königsberg, ETH Zürich, University of Göttingen
Best known for his foundational work on space-time 4D space which facilitated geometric interpretations of Einstein's special theory of relativity (1905).

33

k -Nearest Neighbor: Curse of Dimensionality $\mathbb{R}^n \rightarrow +\infty$

$$\mathbb{R}^n \rightarrow \mathbb{R}: d(\mathbf{x}, \mathbf{y}) = [\sum_{i \in \{1, \dots, n\}} (x_i - y_i)^2]^{\frac{1}{2}}$$



$|\mathbf{O}^*|$ labeled observations $\rightarrow |\mathbf{O}^*|$ distances with unseen $\hat{o}$

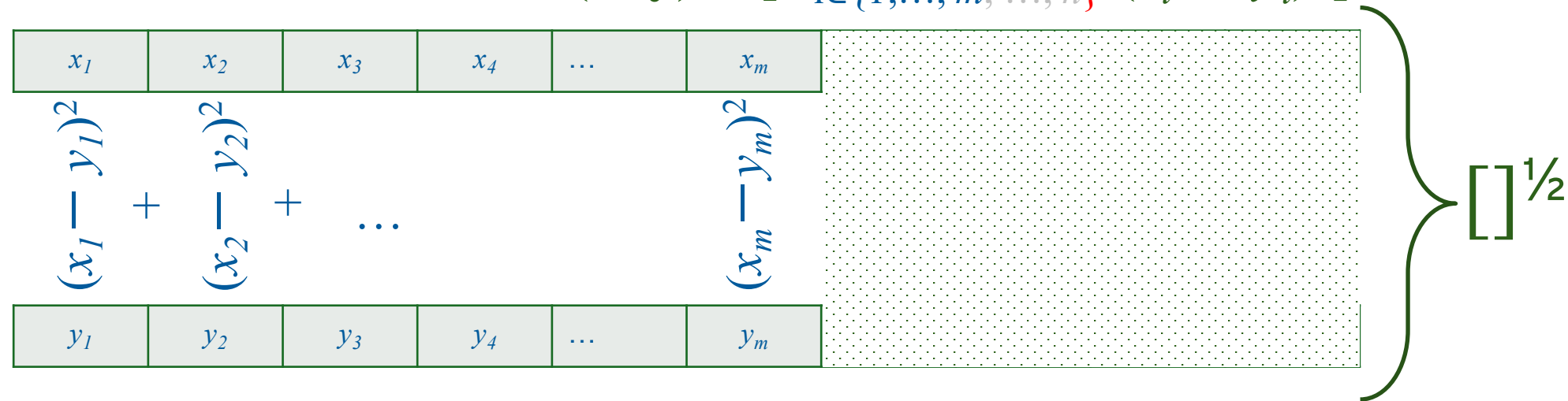
Each distance $\rightarrow n$ (subtractions + squares)

$\rightarrow n \times |\mathbf{O}^*| \rightarrow$ **Linearly grows** for both the size of training set and dimensions

k -Nearest Neighbor: Shadowing (Projection)

Naïve Dimensionality Reduction: Simply take the first few dims

$$\mathbb{R}^{n \gg 0} \rightarrow \mathbb{R}^{m > 0} \rightarrow \mathbb{R}: d(\mathbf{x}, \mathbf{y}) = [\sum_{i \in \{1, \dots, m, \dots, n\}} (x_i - y_i)^2]^{\frac{1}{2}}$$



$|\mathbf{O}^*|$ labeled observations $\rightarrow |\mathbf{O}^*|$ distances with unseen $\hat{o}$

Each distance $\rightarrow m$ (subtractions + squares)

$\rightarrow m \times |\mathbf{O}^*| \rightarrow \text{Manageable}$

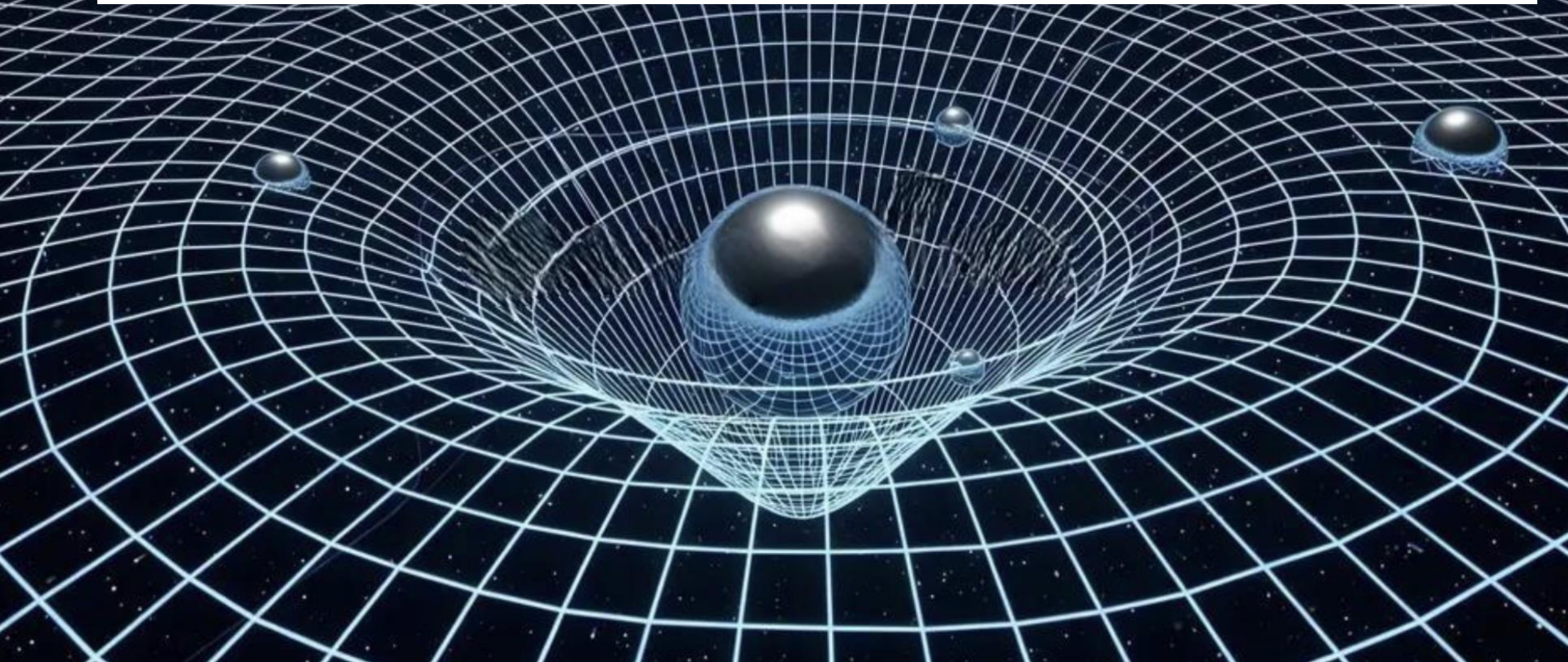


An image of the core region of a supermassive black hole, processed from a sparse array of radio telescopes
https://en.wikipedia.org/wiki/Black_hole

k -Nearest Neighbor: Locally-Sensitive Hashing (LSH)

Hashing Collisions for Efficient Nearest Neighbor Search → Almost Constant (1-3) Steps

CSC502: Systems for Massive Datasets, Mining of Massive Datasets: Ch03: Finding Similar Items



37

k-Nearest Neighbor: Domain Challenges

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT} \{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k \mathbf{1} [c_i = c] \right\}$$


d, math “*kindly*” show us how to measure distances in Euclidian space.

The remaining task is ...

38

k -Nearest Neighbor: Domain Challenges

How to map a real-world observation to a point in Euclidian space?

People

Animals (Birds, ...)

Objects (Houses, Cars, ...)

Image (Biometric Iris)

Audio (Voice, Music)

Video (Movies)

Text (Words, Sentence, Docs)

Timeseries

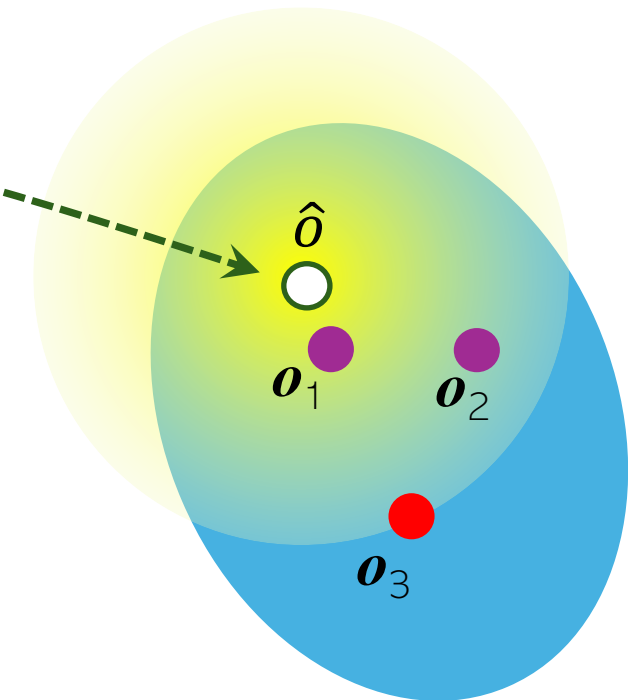
Genes

Spatial (GIS)

Light Bulbs

...

Encoding
Embedding
 $e(o)$



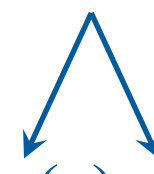
Euclidian Space
(Embedding Space)

39

 k -Nearest Neighbor: Domain Challenges

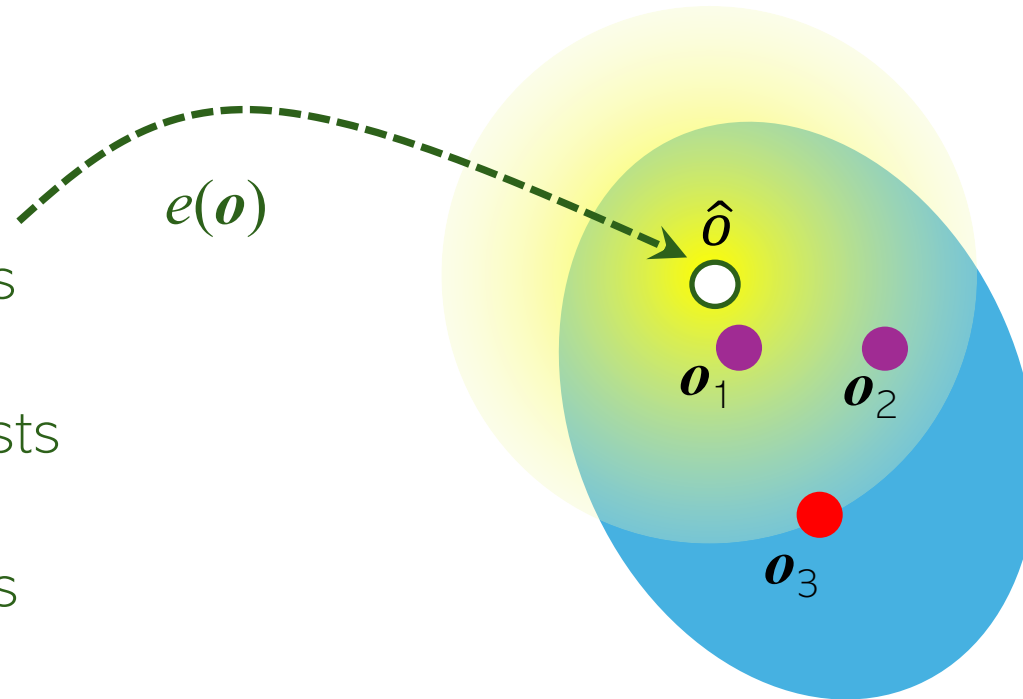
$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$


$$d(\hat{o}, o) \approx d(e(\hat{o}) \in \mathbb{R}^n, e(o) \in \mathbb{R}^n) = d(\mathbf{x}_{\hat{o}}, \mathbf{y}_o) = [\sum_{i \in \{1, \dots, n\}} (x_i - y_i)^2]^{1/2}$$


$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(e(\hat{o}), e(o))\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$


40 k -Nearest Neighbor: Text

Words
Sentences
Docs
Social Posts
Books
Webpages
...



41

k -Nearest Neighbor: Text: Word Space $\rightarrow$ Euclidian Space

Real-World Problem: Named Entity Annotator

Task: Classification $\mathbf{C} = \mathbf{Entities} = \{\text{Person, Organization, Event, Food, Animal}\}$

Approach: Fully Supervised $\mathbf{O}^* = \{(w, c)\}$, $\mathbf{U}^* = \{(\hat{w}, f(\hat{w}))\}$

Technique: $f: k\text{-NN} \rightarrow d(\hat{w}, w) \approx d(e(\hat{w}), e(w)) \rightarrow e: \mathbf{U} = \mathbf{Vocabs} \rightarrow \mathbb{R}^{n=1}$

42

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=1}$

Indexing $\rightarrow$ Enumerate words in a language's dictionary (vocabulary set)



$$e(\hat{w}: \text{'aardvark'})=0$$



$$e(w: \text{'apple'})=1050$$



$$e(w: \text{'zyzzyva'})=|V|$$

Does $e(.)$ preserve “real-world” proximity to “spatial” proximity? **No!**

$$\begin{aligned} |e(\text{'aardvark'}) - e(\text{'apple'})| &< |e(\text{'aardvark'}) - e(\text{'aardvark'})| \\ |0 - 1050| &< |0 - 520,000| \end{aligned}$$

‘aardvark’ is closer to ‘apple’ vs. ‘zyzzyva’!

‘aardvark’ is tech company!